Chapter 2

Proof of Concept: Polymer-Based Composite Organic Memristive Devices

2.1 Introduction and Motivation

This chapter reports the proof-of-concept polymer-electrolyte memristive device announced at the end of **??**. The composite combines a semiconducting poly(phenylene vinylene) (PPV) derivative (*Super Yellow*, SY), an oxygen-rich hyperbranched polyester-amide ion-conducting host (*Hybrane*® DEO750 8500), and a dissociated lithium salt (lithium triflate, LiOTf, $LiCF_3SO_3$), all processed from a common solvent into a standard two-terminal (2-T) sandwich on indium tin oxide (ITO) and silver. The chapter has two purposes within the thesis. First, it tests whether the materials logic of **??** translates into a working device that supports the full set of synaptic functions in a single platform. Second, it isolates the mechanistic ingredient (the cation–host coordination chemistry) that the comparative work of Chapter 3 then tests across cations and compositions.

The literature already contained, at the start of this work, several partial demonstrations of these functions in different architectures, as framed in **??**. The polyaniline-based electrochemical junctions of Erokhin and co-workers [1–3] demonstrated ion-mediated analogue switching, but in a multi-layer geometry rather than a single composite film. The $[Ru(bpy)_3]^{2+}$ light-emitting memristor of Zakhidov and co-workers [4] demonstrated bistable ionic redistribution, but without analogue gradation or full reversibility. The redox-active two-terminal device of Liu and co-workers [5] demonstrated tunable short-term to long-term memory transitions, but relied on partially irreversible redox transformations of the kind documented for redox-based organic memory more broadly [6]. Organic electrochemical transistors achieved reversible analogue control under biologically realistic learning rules [7–10], but at the architectural cost of an additional gate terminal and a contacted electrolyte volume. Each of these implementations covered part of the specification developed in **??** (analogue, reversible, multi-state, low-energy conductance changes set by electrical pulses), but the combination of those features in a single two-terminal organic device with an interpretable ion-mediated mechanism remained, at that point, an open problem.

The active-layer logic adopted here is the composite logic of **??**. The three functional roles — electronic transport, ionic transport, and mobile-charge supply — are assigned to three different components, so that each role can be tuned chemically without redesigning the rest of the stack. *Super Yellow* supplies the electronic-conductance pathway. *Hybrane* supplies an amorphous, low-crystallinity ion-transport host with a high density of hard oxygen donors. LiOTf supplies a hard alkali cation paired with a soft, charge-delocalised anion, so that the cation–host and anion–host interactions are asymmetric in the hard–soft acid–base sense of Pearson [11]. The mechanistic account developed in section 2.5 attributes the two memory regimes to this asymmetry.

The chapter shows that the composite supports, in a single 2-T platform, a full set of synaptic functions that fall into two families. The first family comprises three dynamical measurements: current–voltage hysteresis characteristic of memristive switching, potentiation as a function of the number of applied voltage pulses, and depotentiation as a function of the delay before read-out. These three measurements recur, on the same 2-T device geometry, throughout the comparative work of Chapters 3 and 4; they are introduced first, and as an explicitly named set, so that the reader meets them here once rather than encountering them repeatedly in the later chapters without context. The second family comprises extended synaptic protocols that build on the first and are specific to this chapter: excitatory post-synaptic current (EPSC) responses, demonstrated here with four readily distinguishable readout states; the resolution of the depotentiation measurement into a short-lived and a longer-lived memory regime, denoted short-term memory (STM) and long-term memory (LTM) following the convention of the artificial-synapse literature [12, 13]; and spike-timing dependent plasticity (STDP) with a time constant in the biologically realistic range [14, 15]. The interpretation tying these observations together is developed in section 2.5: the fast component is associated with displacement of the weakly bound triflate anion, while the slow component is associated with displacement of the lithium cation from its hard oxygen coordination environment. The same interpretation yields the cation-substitution prediction tested in Chapter 3.

Within the arc of the thesis the present chapter is narrow in role: it is the only chapter in which the full synaptic suite is treated as the main evidence, while only the three dynamical measurements named above are carried forward to the comparative corpus. The two specific handoffs onto the rest of the thesis are taken up at the end of the chapter.

This chapter is organised as follows. Section 2.2 sets out the molecular composition and the design rationale of the composite. Section 2.3

describes the fabrication protocol and the morphological characterisation of the active layer. Section 2.4 reports the full electrical characterisation suite. Section 2.5 develops the mechanistic interpretation of the two memory regimes in terms of cation–host coordination chemistry, together with the injection-limited and bulk-doping descriptions of the polymer film. Section 2.6 addresses device stability and reproducibility. Section 2.7 contextualises the results against the state of the art at the time of publication. Section 2.8 closes the chapter and sets up the comparative and computational work that follows.

2.2 Materials Design and Composite Formulation

2.2.1 Design Rationale

The operating principle of the device is the same one developed in **??**: ionic migration within a solid-state polymer matrix under an applied electric field redistributes mobile ions near the electrodes, which in turn modifies the energy barriers for charge injection and the doping level of the semiconducting polymer, and therefore the electronic conductance of the device. The composite active layer is built to satisfy three functional requirements simultaneously:

1. **Electronic-conductance pathway:** a semiconducting polymer provides the medium through which the electronic current flows, and whose conductivity is susceptible to modulation by ionic redistribution and electrochemical doping.
2. **Ion-transport medium:** a polymer electrolyte provides a structural environment in which dissolved ions remain mobile under an

applied field, yet are retarded enough by intermolecular interactions that, once the field is removed, the conductance change persists for seconds — long enough to be read out — before fading back toward equilibrium.

3. **Ion source:** a dissociated salt provides the mobile charge carriers (a cation and an anion) whose differential response to the field is proposed to underlie the distinct memory regimes reported in section 2.4.

The composition adopted here is built around a deliberate asymmetry in Lewis acid–base character between the two ions of the salt. LiOTf is chosen because Li^+ is a hard Lewis acid with a high charge-to-radius ratio ($\lambda = z/r \approx 1.3\,\text{Å}^{-1}$–$1.7\,\text{Å}^{-1}$ across the four- to six-fold oxygen coordination that Li^+ adopts in polyether hosts, with r_{Li^+} in the range 0.59 Å–0.76 Å for the corresponding Shannon effective radii [16]), whereas the triflate anion $CF_3SO_3{}^-$ is a soft Lewis base with a diffuse, low-charge-density anionic centre. The ion-transport polymer, *Hybrane*, contains a high density of ether and ester oxygen atoms that act as hard Lewis bases. Under the hard–soft acid–base (HSAB) reasoning of Pearson [11] already used in **??**, hard acids associate preferentially with hard bases. The Li^+ cations are therefore tightly coordinated by the oxygen atoms of *Hybrane*, while the $CF_3SO_3{}^-$ anions interact only weakly with the host. Under bias, the weakly coordinated triflate anions are displaced at low voltage, while the strongly coordinated lithium cations require a substantially higher field to break free of their coordination environment. This differential mobility is proposed as the molecular origin of the two memory regimes (short-term memory and long-term memory) reported below.

2.2.2 Super Yellow: The Semiconducting Polymer

The semiconducting component of the composite is *Super Yellow*, the trivial name for the high-molecular-weight PPV copolymer poly[{2,5-di(3′,7′-dimethyloctyloxy)-1,4-phenylenevinylene}-*co*-{3-(4′-(3″,7″-dimethyloctyloxy)phenyl)-1,4-phenylenevinylene}-*co*-{3-(3′-(3′,7′-dimethyloctyloxy)phenyl)-1,4-phenylenevinylene (commercial name: PDY-132, Merck). The chemical structure features a π-conjugated backbone of alternating phenylene and vinylene units with branched alkoxy side chains that confer solubility in common organic solvents and reduce intermolecular packing to mitigate crystallisation-induced phase separation in blends [17, 18].

The frontier orbital levels of *Super Yellow* ($E_{\mathrm{HOMO}} \approx -5.4\,\mathrm{eV}$, $E_{\mathrm{LUMO}} \approx -3.0\,\mathrm{eV}$) give a HOMO–LUMO gap of approximately 2.4 eV [18], consistent with the yellow emission under photoexcitation and electroluminescence that underlies its application in organic light-emitting devices [19]. In the context of memristive applications, three properties are relevant:

1. **Hole mobility:** adequate hole transport through the bulk of the film provides the baseline electronic-conductance pathway.

2. **Electrochemical doping:** susceptibility to p-doping by accumulation of cations at the polymer chains (or n-doping by anion accumulation) modifies the conductance under the electrochemical-doping mechanism.

3. **Solution processability:** solubility in cyclohexanone makes it compatible with the other components of the composite.

The suitability of *Super Yellow* in ion-containing composite systems has been validated previously in the context of light-emitting electrochemical cells (LECs) [20], where films incorporating *Super Yellow* together with Hybrane and lithium salt demonstrated electroluminescence lifetimes exceeding 1600 h, confirming the photochemical and electrochemical stability of the blend under long-term operation.

2.2.3 Hybrane DEO750 8500: The Ion-Transport Polymer

Hybrane is a hyperbranched polyester-amide synthesised by the polymerisation of a bis-2-oxazoline monomer with a dicarboxylic acid, produced by Polymer Factory (Sweden) [21]. The hyperbranched architecture generates a highly irregular, amorphous macromolecular structure with a very large number of chain ends. These chain ends, together with the ether and ester oxygens distributed throughout the backbone, provide a dense array of hard Lewis base coordination sites capable of solvating alkali metal cations.

Two properties of *Hybrane* make it well suited as the ion-conducting component of this composite. First, its glass transition temperature T_g is sufficiently low that appreciable segmental mobility of the polymer chains is maintained at room temperature, which enables the segmental-motion-assisted ionic transport described in **??** [22]. Second, the hyperbranched architecture suppresses crystallisation: in contrast to linear semicrystalline polymers such as poly(ethylene oxide) (PEO), *Hybrane* remains amorphous, yielding a homogeneous matrix throughout the film. Crystalline domains in PEO-based solid electrolytes are known to impede ionic transport [23, 24], and the amorphous host therefore avoids that limitation at the proof-of-concept stage of the thesis.

The ion-transport mechanism in *Hybrane* is primarily segmental: as polymer chain segments undergo local conformational relaxation, coordinated cations are passed from one oxygen coordination site to the next along the segmental motion pathway. Because this motion is coupled to the segmental dynamics of the polymer host above T_g, the transport follows a non-Arrhenius (super-Arrhenius) temperature dependence of the Williams–Landel–Ferry (WLF) type rather than a simple activated law [25]. At room temperature, the apparent activation energy for ion hopping in such systems is of order $50\,\mathrm{kJ\,mol^{-1}}$ [26], consistent with the observation that relatively modest applied voltages (1 V–3 V) are sufficient to displace ions over short distances (sub-µm) within the thin-film device.

2.2.4 Lithium Triflate: The Ionic Salt

LiOTf (lithium trifluoromethanesulfonate) is a well-characterised solid electrolyte salt widely employed in polymer batteries and LECs [27, 28]. Its choice for this system is motivated by three considerations:

1. **Processing compatibility:** it dissolves readily in cyclohexanone and blends homogeneously with *Super Yellow* and *Hybrane* without phase separation.

2. **Hard cation:** the lithium cation is a hard Lewis acid with a small ionic radius (0.59 Å–0.76 Å for four- to six-fold oxygen coordination [16]) and a formal charge of +1, giving a charge density well matched to the hard Lewis base oxygen atoms of *Hybrane*.

3. **Soft anion:** the triflate anion has a large, diffuse anionic centre with the negative charge delocalised over the three electronegative fluorine atoms and the sulfonyl oxygen atoms, rendering it a soft Lewis base with low affinity for the Hybrane oxygen coordination sites.

The dissociation of LiOTf in the composite is enhanced by the solvating ability of the ethylene oxide segments of *Hybrane*, analogous to the solvation of lithium salts in PEO-based electrolytes [22]. The equilibrium ionic conductivity of the composite is governed by the ion pair dissociation equilibrium, the number density of free charge carriers, and their individual mobilities within the matrix. In the memristive context, however, the relevant quantity is not the equilibrium conductivity of the composite, but the ability to reversibly redistribute the ions on the timescale of the applied voltage pulses — a kinetic, rather than thermodynamic, property.

2.2.5 Composite Formulation and Preparation

The three components were dissolved separately in cyclohexanone at initial concentrations of $15\,\mathrm{mg\,mL^{-1}}$ (*Super Yellow*), $10\,\mathrm{mg\,mL^{-1}}$ (*Hybrane*), and $5\,\mathrm{mg\,mL^{-1}}$ (LiOTf). The *Super Yellow* solution was stirred continuously at 50 °C overnight to ensure complete dissolution of the high-molecular-weight polymer. The three solutions were then combined in a volumetric ratio that results in a mass ratio of *Super Yellow*:*Hybrane*:LiOTf = 1:0.30:0.09. After mixing, the final concentrations are $8.72\,\mathrm{mg\,mL^{-1}}$ (*Super Yellow*), $2.62\,\mathrm{mg\,mL^{-1}}$ (*Hybrane*), and $0.78\,\mathrm{mg\,mL^{-1}}$ (LiOTf). All solution handling was performed under a dry nitrogen atmosphere to prevent moisture absorption, which would introduce uncontrolled ionic species and degrade reproducibility.

The compositional ratios were arrived at through systematic optimisation [17]. Increasing the relative concentration of LiOTf beyond the chosen ratio was found to produce phase separation detectable by optical microscopy and a deterioration in film homogeneity. Conversely, reducing the salt content below the chosen ratio produced devices with insufficient ionic carrier density to generate well-defined conductance states. The chosen ratio represents a balance between maximising ionic carrier concentration and maintaining film morphological integrity.

This composite is identical in formulation to the active layer of the LECs studied by Mardegan *et al.* [20], where its electroluminescent performance and long operational lifetime (>1600 h) were established; the present work redeploys the same architecture for memristive operation.

2.3 Device Architecture and Fabrication

2.3.1 Device Geometry: The Two-Terminal Vertical Structure

The device adopts a vertical sandwich geometry, in which the active layer is placed between a bottom transparent electrode and a top metallic electrode (see fig. 2.1). This 2-T architecture is the simplest memristive configuration available: it contains no gate electrode and requires no transistor to address the device, in contrast to the *three-terminal* (3-T) organic synaptic transistors discussed in **??** [7, 8, 29].

The 2-T choice matters for two reasons. First, it reduces fabrication complexity, given that the device is made by a sequence of only three depositions (ITO substrate, spin-coated active layer, evaporated metal), which makes the same stack compatible with the comparative corpus of Chapter 3. Second, a 2-T device that shows coexisting short-term memory and long-term memory without an independent gating signal shows that the memory physics is intrinsic to the active material, rather than being supplied externally by a gate field acting on a separate channel.

The bottom electrode is a pre-patterned layer of ITO deposited on borosilicate glass. ITO combines optical transparency with adequate electronic conductance ($\sigma \approx 10^4\,\mathrm{S\,cm^{-1}}$) [30], allowing optical inspection of the active layer and characterisation of the device stack by techniques such as UV-Vis spectroscopy and electroluminescence imaging. The work function of ITO ($\Phi_{\mathrm{ITO}} \approx 4.7\,\mathrm{eV}$) is well matched to the ionisation potential of PPV-family semiconductors [31], facilitating hole injection into the active layer.

[Figure placeholder]
Two-terminal device architecture and proposed STM/LTM mechanism
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Figure 2.1: Schematic of the two-terminal vertical synaptic device. (a) The biological synapse used as template. (b) Device stack: ITO-glass bottom electrode / composite active layer / Ag top electrode (100 nm). (c) Molecular structures of the three active-layer components: *Super Yellow* (semiconducting PPV copolymer), *Hybrane* (hyperbranched polyester-amide ion-transport host), and LiOTf ($LiCF_3SO_3$). (d) Proposed mechanism: at low voltage the fast triflate anions redistribute (STM); at high voltage the slow Li^+ cations are also displaced (LTM). Adapted from Ref. [17].

The top electrode is silver (Ag), deposited by thermal evaporation. Ag was selected over gold for reasons of cost and practical feasibility; its work function ($\Phi_{\mathrm{Ag}} \approx 4.3\,\mathrm{eV}$) creates a modest electron injection barrier with the *Super Yellow* conduction band, consistent with asymmetric injection conditions often observed in LEC-type structures [32]. The junction active area defined by the shadow mask geometry is $0.0825\,\mathrm{cm}^2$.

2.3.2 Substrate Preparation

ITO-patterned glass substrates were subjected to a sequential ultrasonic cleaning procedure prior to device fabrication. The substrates were ultrasonicated successively in aqueous Mucasol detergent solution, deionised water, and 2-propanol, each for 15 min. This procedure removes organic contaminants and metallic particulates from the ITO surface, both of which would increase surface roughness and create pinhole defects in the subsequently deposited active layer. After ultrasonic cleaning, the substrates were dried under a flow of dry nitrogen gas and transferred immediately into a nitrogen-filled glovebox (oxygen and water levels < 1 ppm), where all subsequent processing steps were performed.

Surface contamination of the ITO is a critical concern in the fabrication of thin-film organic devices: even sub-nanometre organic residues on the substrate surface can disrupt the wetting behaviour of the solution during spin-coating, leading to dewetting instabilities, pinholes, and non-uniform coverage. The Mucasol cleaning step is particularly effective at removing hydrocarbon contaminants, and the final 2-propanol rinse removes any residual Mucasol surfactant.

2.3.3 Active Layer Deposition

The composite solution (prepared as described in section 2.2.5) was deposited onto the cleaned ITO-glass substrates by spin-coating at 2000 rpm for 60 s. The spin speed was selected on the basis of a systematic optimisation study in which spin speeds ranging from 500 rpm to 3000 rpm were investigated (see Supporting Information of Ref. [17]). At lower spin speeds, the resulting films were too thick and exhibited non-uniform coverage; at higher speeds, film thickness was reduced to the point where statistical roughness from the substrate became a significant fraction of the total film thickness, compromising device reproducibility.

Following deposition, the coated substrates were transferred to a hotplate within the glovebox and annealed at 75 °C for 3 h. The annealing step serves three purposes:

1. **Solvent removal:** it removes residual cyclohexanone solvent, which would otherwise act as a plasticiser and modify the ionic mobility of the matrix.

2. **Chain relaxation:** it allows the polymer chains to relax into a morphologically stable configuration.

3. **Component mixing:** it promotes intimate mixing of the three components at the molecular scale, maximising the number of Li^+ coordination sites accessible to the triflate anion within the Hybrane matrix.

2.3.4 Morphological Characterisation of the Active Layer

Film Thickness by Profilometry

The thickness of the deposited active layer was determined by contact profilometry using an Ambios XP-1 profilometer. A step edge was created by masking part of the substrate during deposition and measuring the height difference between the coated and uncoated regions. The measured thickness at 2000 rpm spin speed is 209 nm. This value is typical for single-layer active materials in organic electronic devices and represents a balance between: (i) providing sufficient ionic carrier density in a three-dimensional bulk (proportional to film thickness at constant volume concentration); and (ii) maintaining low series resistance in the electronic conduction channel (resistance scales with thickness for ohmic conduction).

Surface Topography by Atomic Force Microscopy

The surface topography of the active layer was characterised by atomic force microscopy (AFM) in tapping mode using a Bruker Dimension

Icon instrument. A silicon probe with resonant frequency 300 kHz and force constant $40\,\mathrm{N\,m^{-1}}$ was employed at a scan rate of 0.6 Hz. Films deposited at several spin speeds were imaged to assess the relationship between processing conditions and surface morphology (see Supporting Information of Ref. [17]).

The AFM images reveal a smooth, uniform surface with root-mean-square (RMS) roughness in the range of 1 nm–3 nm across the range of spin speeds investigated [17]. The absence of large-scale phase-separated domains or crystalline features confirms that the three-component composite remains homogeneously blended in the solid state, consistent with the role of *Hybrane* in suppressing salt crystallisation. The low roughness is important for device performance: rough surfaces create local field enhancements at protrusions that can cause premature electrical breakdown during characterisation, and they also increase the statistical spread of active layer thickness across the device area, which would broaden the distribution of switching voltages.

2.3.5 Top Electrode Deposition

After morphological characterisation, the top Ag electrode was deposited by thermal evaporation in a dedicated evaporation system housed within the glovebox. A shadow mask defining the $0.0825\,\mathrm{cm^2}$ junction area was placed in contact with the active layer surface. Ag was evaporated at a deposition rate of $0.1\,\mathrm{nm\,s^{-1}}$–$0.5\,\mathrm{nm\,s^{-1}}$ to a total thickness of 100 nm, monitored in real time by a quartz crystal microbalance. The slow initial evaporation rate was employed to minimise the kinetic energy of impinging Ag atoms and thereby prevent penetration of the metal into the soft polymer film, which would create metallic filaments and short-circuit the device [33].

The completed device was stored under nitrogen atmosphere until measurement. The devices were functional without encapsulation in ambient air for the duration of most experiments; degradation studies (section 2.6) confirm a functional lifetime of approximately two weeks (300 h) under ambient conditions.

2.4 Electrical Characterisation

All electrical measurements were performed using a Keithley 2450 SourceMeter connected to a custom-built probe station. Measurement protocols were implemented in the Keithley TestScript Builder (TSB) programming environment. All measurements were carried out under a nitrogen atmosphere unless otherwise stated (degradation studies explicitly employed ambient air conditions). Throughout this section, voltage is applied to the Ag top electrode relative to the grounded ITO bottom electrode.

The characterisation is organised into two groups, following the distinction drawn in section 2.1. Sections 2.4.1 to 2.4.3 report the three dynamical measurements that are carried forward to the comparative corpus of Chapters 3 and 4 — current–voltage hysteresis, variable-number-of-pulses potentiation, and variable-delay depotentiation. Sections 2.4.4 and 2.4.5 then report the extended synaptic protocols — the EPSC multi-state readout and spike-timing dependent plasticity — which are specific to this device and are not propagated to the later chapters. The short- and long-term memory regimes (short-term memory/long-term memory) are obtained as an interpretation of the variable-delay depotentiation measurement, and are therefore presented together with it in section 2.4.3.

2.4.1 Current-Voltage Hysteresis: Signature of Memristive Behaviour

The fundamental electrical fingerprint of a memristive device is a current-voltage (I–V) hysteresis loop with a characteristic pinched appearance at zero voltage [34–36]. This shape arises because, at $V = 0$, the net current is zero regardless of the history of the device (a memristor cannot spontaneously drive current), yet the *slope* of the I–V curve at zero voltage (i.e., the instantaneous conductance) depends on the history of applied voltage.

Figure 2.2 shows the I–V curves obtained by applying successive triangular voltage sweeps to the device at a sweep rate of $0.25\,\mathrm{V\,s^{-1}}$, with the voltage ramped from 0 V to a maximum of 1.2 V and back to 0 V (single polarity, positive). A waiting period of 10 s was observed between successive sweeps to allow partial ionic relaxation. Ten sequential sweeps are shown.

Several features of these curves are noteworthy:

Progressive increase of conductance. The current at any fixed voltage *increases* with each successive sweep. From the first to the second cycle, the current at 1 V increases by 140 %. By the tenth cycle, the incremental increase from cycle 9 to cycle 10 is 6.3 % — suggesting that the system is approaching a saturation conductance level asymptotically. This behaviour reflects the progressive redistribution of ions toward the electrodes with each sweep: each successive cycle leaves the ions in a slightly more asymmetric configuration than the previous one, producing a slightly lower resistance state.

Hysteresis without current rectification. In all cycles, the return sweep (decreasing voltage) produces lower current than the forward sweep (increasing voltage) at the same voltage, generating the characteristic

[Figure placeholder]
Current–voltage hysteresis over ten successive sweeps
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Figure 2.2: Current–voltage hysteresis of the composite memristive device. Ten successive positive triangular sweeps (0 V–1.2 V, $0.25\,\mathrm{V\,s^{-1}}$, 10 s between cycles) show a progressive increase of conductance; the current at 1 V rises by 140 % from cycle 1 to cycle 2. The clockwise loop closes at the origin (pinched hysteresis) without abrupt switching. Data from Ref. [17].

clockwise hysteresis loop for positive voltage sweeps. This is distinct from simple rectification (diode behaviour): a diode would produce the same forward and reverse I–V curves; the hysteresis here reflects a time-dependent internal state variable (the ionic configuration) that lags the applied voltage.

Absence of abrupt switching. Unlike digital resistance switching (SET/RESET transitions), in which the resistance changes abruptly by orders of magnitude at a threshold voltage, the conductance here evolves continuously and smoothly with each cycle. This *analogue* gradation matters for the synaptic role outlined in **??**, since it allows the device state to be tuned to one of many distinguishable intermediate levels rather than being confined to two (or a few) binary states.

Behaviour under negative polarity. Application of negative voltage sweeps (not shown individually in the main figure) produces a qualitatively different hysteresis: the current decreases rather than increases with successive negative cycles. This has been attributed to electrochemical side-reactions in the vicinity of the ITO electrode under large negative bias, which generate a thin insulating layer at the ITO/active layer interface analogous to the behaviour reported in MEH-PPV-based LECs [28, 37]. This effect is not detrimental to memristive operation when devices are used in the single-polarity positive regime.

2.4.2 Variable-Number-of-Pulses Potentiation

Potentiation by a train of write pulses is the second of the three dynamical measurements introduced at the start of this section. Two complementary experiments are reported: a reversible potentiation/depression cycle that establishes the analogue, multi-state character of the conductance, and a measurement of the conductance change as a function of the number of applied pulses.

Reversible Potentiation and Depression

The ability to set the device conductance to a specific value by applying a controlled number of voltage pulses — and to subsequently decrease it back toward the initial value by pulses of opposite polarity — is demonstrated by the potentiation/depression experiment shown in fig. 2.3.

A pulsed voltage protocol was applied: 50 consecutive write pulses of 1 V amplitude and 0.1 s duration, followed by 50 write pulses of −2 V. The conductance was monitored at the peak of each pulse (i.e., at the 1 V and −2 V levels, respectively). This protocol is analogous to the repeated presynaptic stimulation of a biological synapse and its subsequent inhibition.

The results demonstrate that the conductance can be continuously potentiated from its initial value by more than 200 % during the 1 V pulse train, and subsequently depotentiated back toward the initial value during the −2 V train. The potentiation and depression curves are smooth and monotonic, without any abrupt transitions. The full reversibility of the process — the conductance returns to a value near the initial level after the complete depression protocol — confirms that the underlying ionic redistribution is non-destructive and does not permanently alter the chemical composition or morphology of the active layer.

[Figure placeholder]
Reversible potentiation/depression under a pulse train
to be generated from data — see
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Figure 2.3: Analogue multi-state conductance tuning under a pulse-train protocol. Fifty consecutive 1 V write pulses continuously potentiate the conductance by more than 200 %; fifty subsequent −2 V pulses depotentiate it back toward the initial state. The smooth, monotonic trajectories demonstrate reversible analogue tuning without abrupt SET/RESET transitions. Data from Ref. [17].

The energy cost of each synaptic event (one write pulse of 1 V, 0.1 s) can be estimated from the average current during the pulse. The resulting energy per event is approximately 50 nJ, or approximately 6 fJ μm^{-2} of device area (the 0.0825 cm^2 junction). This value is consistent with those reported for organic synaptic devices operating in the picojoule-to-nanojoule range [38], and comparable to the picojoule-to-nanojoule switching energies reported for inorganic memristive devices [39].

Quantitative Pulsed Read–Write Protocol

To quantify the potentiation as a function of pulse number, and to separate the act of writing from the act of reading, a non-perturbing read protocol was adopted; the same protocol is reused for the depotentiation measurement of section 2.4.3. The read–write–wait sequence, shown schematically in fig. 2.4, comprises three steps:

1. **Read.** A reading pulse ($V_{\mathrm{read}} = 0.5\,\mathrm{V}$) is applied to establish the initial conductance G_0. This reading voltage is intentionally low: the activation threshold for ionic redistribution in this system was determined by independent experiments to lie above $0.7\,\mathrm{V}$ [17], so $V_{\mathrm{read}} = 0.5\,\mathrm{V}$ measures the conductance state without perturbing it.

2. **Write.** A train of N_{pulses} write pulses (V_{write}, duration $0.1\,\mathrm{s}$ each) is applied.

3. **Wait and read.** After a waiting time t_{wait} under open-circuit (high-impedance) conditions, a second reading pulse is applied to measure the final conductance G_f.

The conductance change is reported as the percentage ratio $\Delta G = (G_f/G_0) \times 100$.

Between experiments, the device is fully reset by connecting the junction to a short circuit for $300\,\mathrm{s}$, which discharges the ionic double layers and restores the homogeneous equilibrium ion distribution. The suitability of this reset protocol was verified by confirming that G_0 returns to its original value within the measurement uncertainty before each new experiment.

[Figure placeholder]
Read–write–wait protocol schematic
to be generated from data — see
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Figure 2.4: Pulsed read–write–wait protocol used for the potentiation and retention measurements. A low read pulse ($V_{\text{read}} = 0.5\,\text{V}$) establishes the baseline conductance G_0; a train of N_{pulses} write pulses perturbs the ionic configuration; after a controlled waiting time t_{wait} a second read pulse measures the relaxed state G_f. Adapted from Ref. [17].

Potentiation as a Function of Pulse Number

Figure 2.5 shows the dependence of ΔG on N_{pulses} for $V_{\text{write}} = 1\,\text{V}$ and $t_{\text{wait}} = 150\,\text{ms}$. The conductance ratio increases monotonically with N_{pulses}, reaching values in excess of $10^3\%$ for $N_{\text{pulses}} > 10^3$. The relationship is approximately linear in log-log space for $N_{\text{pulses}} < 100$ and saturates at large pulse numbers, consistent with a scenario in which the degree of ionic asymmetry that can be built up under a given write voltage is bounded by the total number of mobile ionic carriers.

The device-to-device reproducibility is high: the shaded error band in fig. 2.5 represents the standard deviation across measurements on multiple independent devices, and the relative standard deviation is below $10\,\%$ for $N_{\text{pulses}} < 100$. The dispersion increases modestly at large pulse numbers, reflecting the probabilistic nature of the ionic migration dynamics at large driving forces.

[Figure placeholder]
Potentiation versus number of write pulses
to be generated from data — see
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Figure 2.5: Dependence of the conductance change $\Delta G = (G_f/G_0)\times 100\%$ on the number of write pulses N_{pulses} for $V_{\text{write}} = 1\,\text{V}$ and $t_{\text{wait}} = 150\,\text{ms}$. The monotonic increase over more than three decades of pulse count demonstrates analogue potentiation. The shaded band is the standard deviation across independent devices. Data from Ref. [17].

2.4.3 Variable-Delay Depotentiation: Short- and Long-Term Retention

The third dynamical measurement carried through the thesis probes how a written conductance state fades once the drive is removed: the depotentiation of the device measured as a function of the delay before read-out. This is one of the most important characterisation protocols for neuromorphic devices, because it determines whether the device can emulate short-lived and longer-lived synaptic states on experimentally accessible timescales. In the artificial-synapse literature these regimes are commonly labelled short-term memory and long-term memory; in the present system, the longer-lived regime remains on the sub-minute scale and should therefore be understood operationally rather than as a claim of indefinitely persistent storage.

The measurement reuses the pulsed read–write protocol of section 2.4.2, with the waiting time t_{wait} now the swept variable and the pulse number held fixed. Figure 2.6 shows the dependence of ΔG on t_{wait} for three excitation conditions: (i) $V_{\text{write}} = 1\,\text{V}$, $N_{\text{pulses}} = 10$ (STM, blue); (ii) $V_{\text{write}} = 1\,\text{V}$, $N_{\text{pulses}} = 50$ (STM, red); and (iii) $V_{\text{write}} = 3\,\text{V}$, $N_{\text{pulses}} = 10$ (LTM, black).

[Figure placeholder]
Retention kinetics: STM and LTM relaxation curves
to be generated from data — see
`handouts/03_chapter2_figures_needed.md`

Figure 2.6: Memory retention kinetics. Conductance ratio $(G_f/G_0) \times 100\%$ versus waiting time t_{wait} for STM conditions (1 V; 10 and 50 pulses, blue/red) and the longer-lived regime denoted LTM (3 V, 10 pulses, black). Solid lines are fits to the stretched-exponential (Kohlrausch) form of eq. (2.1); fitted characteristic times $\tau_S = 2.5\,\text{s}$–$3\,\text{s}$ (STM) and $\tau_L = 4.7\,\text{s}$ (LTM). Data from Ref. [17].

For short waiting times ($t_{\text{wait}} < 500\,\text{ms}$ for STM conditions, $t_{\text{wait}} < 2\,\text{s}$ for LTM conditions), the conductance first *increases* slightly relative to the value immediately after the write pulse train. This counterintuitive behaviour is *not* ascribed to ionic inertia: ion transport in the polymer matrix is strongly overdamped, so any momentum imparted by the field relaxes on a timescale orders of magnitude shorter than the observed enhancement. It is instead attributed to the continued redistribution of ions after the drive is removed. At the end of the pulse train the ionic configuration and its self-consistent space-charge field have not yet reached their quasi-steady profile; the residual internal field, together with the slow advance of the electrochemically doped region near the electrode, continues to raise the conductance for a brief interval before the diffusion-driven relaxation sets in. An analogous transient enhancement following a stimulus train has been described as “post-tetanic potentiation” in biological synaptic preparations [40, 41].

For longer waiting times, the expected exponential relaxation of conductance is observed as the ionic configuration relaxes back toward equilibrium by thermal diffusion. The relaxation curves were fitted to a modified Kohlrausch (stretched exponential) function [42]:

$$\Delta G(t_{\text{wait}}) = \Delta G_0 \exp\left[-\left(\frac{t_{\text{wait}}}{\tau}\right)^{\beta}\right] + \Delta G_\infty \tag{2.1}$$

where τ is the characteristic relaxation time, β is the stretch exponent ($0 < \beta \leq 1$), ΔG_0 is the initial conductance change, and ΔG_∞ is the residual (non-relaxing) conductance change. The stretched exponential form arises naturally when the relaxation involves a distribution of characteristic times rather than a single time constant [42] — physically, this reflects the heterogeneous environment within the polymer matrix, in which individual ions experience a range of local coordination environments and hence a distribution of activation barriers for migration.

The fitted characteristic times are $\tau_S = 2.5\,\text{s}$–$3.0\,\text{s}$ for STM conditions and $\tau_L = 4.7\,\text{s}$ for LTM conditions. The retention time — defined operationally as the waiting time at which G_f/G_0 returns to within 5 % of the baseline (i.e., $G_f/G_0 < 1.05$) — is 10 s–15 s for STM and exceeds 45 s for the longer-lived regime labelled LTM. These values are directly relevant to the timescales of spike-based computation: the 10 s–15 s STM window is sufficient to maintain working memory over a series of sequential spike inputs, while the $> 45\,\text{s}$ LTM window shows that the device can preserve a write history over many subsequent events without approaching the retention regime of a conventional non-volatile memory.

The molecular mechanism underlying the STM/LTM distinction is discussed in detail in section 2.5.

2.4.4 Excitatory Post-Synaptic Current

The EPSC measurement is a standard protocol in the field of artificial synaptic devices, designed to quantify the number and stability of distinguishable conductance states accessible to the device [12, 13]. The protocol employed here closely mimics the biological experiment in which a patch-clamped neuron is subjected to a defined sequence of pre-synaptic inputs and the post-synaptic current is monitored.

The measurement protocol is as follows (fig. 2.7):

1. **Initialisation.** The device is initialised to its ground state S_0 by applying a fixed read-out bias $V_0 = 1\,\mathrm{V}$ for $1\,\mathrm{s}$. This bias differs in purpose from the sub-threshold probe ($V_{\mathrm{read}} = 0.5\,\mathrm{V}$) used for the non-perturbing read-out of sections 2.4.2 and 2.4.3: the EPSC states are defined operationally by the current response at a single common read-out bias against which all states S_0–S_4 are referenced, not by a passive probe required to leave the state untouched.

2. **Excitation.** A short positive write pulse of $2\,\mathrm{V}$ for $50\,\mathrm{ms}$ is applied; the device is subsequently held at $1\,\mathrm{V}$ for $1\,\mathrm{s}$, during which the conductance spontaneously relaxes to a new excited steady state S_1. The same sequence is repeated to produce a second excited state S_2.

3. **Inhibition.** Two negative write pulses of $-2.5\,\mathrm{V}$ for $50\,\mathrm{ms}$ are applied, producing inhibitory states S_3 and S_4.

The four excited states are readily distinguishable from each other and from the ground state. Defining the EPSC readout ratio $R_n = I_{S_n}/I_{S_0}$ as

[Figure placeholder]
EPSC protocol and four distinguishable readout states
to be generated from data — see
`handouts/03_chapter2_figures_needed.md`

Figure 2.7: Excitatory post-synaptic current (EPSC) measurement. Applied voltage and measured current versus time: the ground state S_0 is set at 1 V; 2 V pulses produce excited states S_1 and S_2; -2.5 V pulses produce inhibitory states S_3 and S_4. The four states are readily distinguishable from the ground state. Data from Ref. [17].

the signed current response of state S_n normalised to that of the ground state S_0 under the fixed read-out bias used in the experiment, the values obtained are: $R_1 = 7.33$, $R_2 = 15.55$, $R_3 = -3.72$, $R_4 = -16.97$. (The ratio is formed from signed currents rather than conductances, so the inhibitory states carry a negative sign; conductance itself remains non-negative throughout.) The large absolute ratios (up to $\sim$16-fold increase or $\sim$17-fold decrease relative to S_0) provide a wide margin for reliable state identification in a practical device, even in the presence of noise. The four-state sequence used here is a representative demonstration rather than an upper bound: given the continuous, analogue conductance tunability established in section 2.4.2, the number of distinguishable readout states is not intrinsically limited to four, and many more intermediate levels could in principle be resolved by the same protocol within the available margin.

The states S_1 and S_2 are reached by *excitatory* (positive polarity) stimulation, while S_3 and S_4 are reached by *inhibitory* (negative polarity) stimulation, in direct analogy with long-term potentiation (LTP) and long-term depression (LTD) at biological synapses. The coexistence of both excitatory and inhibitory states in the same 2-T device, without an independent gate terminal, distinguishes the present platform from the 3-T organic implementations discussed in **??**.

2.4.5 Spike-Timing Dependent Plasticity

Spike-timing dependent plasticity is a timing-dependent Hebbian learning rule: it instantiates Hebb's local correlation principle [43] in a form that was later observed experimentally in biological preparations [14, 44]. It states that the change in synaptic weight depends on the relative timing of pre- and postsynaptic action potentials: if the presynaptic neuron fires shortly *before* the postsynaptic neuron (causal ordering, $\Delta t < 0$), the synapse is potentiated (LTP); if the presynaptic neuron fires shortly *after* the postsynaptic neuron (anti-causal ordering, $\Delta t > 0$), the synapse is depressed (LTD). The magnitude of the change decays exponentially with $|\Delta t|$. This temporal asymmetry is essential for the synaptic encoding of causal relationships in neural circuits and underlies the ability of STDP-based learning rules to implement unsupervised feature extraction and pattern recognition [45].

Experimental Protocol

The STDP experiment requires the emulation of pre- and postsynaptic spike waveforms by voltage pulses applied at the two electrodes of the device. In a 2-T device, both pulses must be applied at the same physical terminal (the Ag electrode), so the experimentally measured total voltage $V(t)$ is the algebraic sum of the presynaptic and postsynaptic voltage waveforms separated by the delay Δt:

$$V(t) = V_{\mathrm{pre}}(t) + V_{\mathrm{post}}(t + \Delta t) \tag{2.2}$$

Each spike waveform consists of a unipolar triangular pulse with maximum amplitude V_{sp} and duration 1.2 s. The delay Δt was varied from -600 ms to 600 ms. The amplitude $V_{sp} = 1.55$ V was chosen to be large enough to activate ionic migration (above the threshold of approximately 0.7 V) while remaining below the onset of electrochemical degradation of the active layer (conservatively estimated at 2 V for the pristine device).

For $\Delta t < 0$ (pre fires before post), the two spike waveforms partially overlap in time with additive polarity, producing a composite pulse whose peak voltage exceeds V_{sp} and which is directed in the positive sense. The ions then redistribute toward the ITO electrode, the conductance increases, and the device records a net synaptic potentiation.

For $\Delta t > 0$ (post fires before pre), the overlap produces an anti-causal summation. Depending on the exact waveforms, the effective net voltage can be of reduced amplitude or of reversed polarity relative to the causal case, so the ionic redistribution is either weaker or reversed and the conductance decreases, in the form of a net synaptic depression.

STDP Curve and Fitting

Figure 2.8 shows the measured STDP function: the percentage conductance change ΔG plotted against the timing delay Δt. The characteristic anti-Hebbian asymmetry is observed: negative Δt (causal) produces positive ΔG (potentiation), and positive Δt (anti-causal) produces negative ΔG (depression). Each branch (potentiation for $\Delta t < 0$, depression for $\Delta t > 0$) is fitted separately, as a function of the absolute delay $|\Delta t|$, by a decaying exponential:

$$\Delta G(\Delta t) = A_{\pm} \exp\left(-\frac{|\Delta t|}{\tau_{\pm}}\right) + G_0 \tag{2.3}$$

where $A_{\pm}$ and $\tau_{\pm}$ are the branch-specific amplitude and time constant ($+$ for the potentiation branch, $-$ for the depression branch) and G_0 is a constant offset. The best-fit values of $\tau_{\pm}$ for both the potentiation and depression branches lie in the range 85 ms–90 ms. This is in close quantitative agreement with the characteristic timescale of biological STDP, which is approximately 100 ms [14, 15]. The device, therefore, not only qualitatively replicates the STDP function but does so with biologically realistic time constants.

[Figure placeholder]
Spike-timing dependent plasticity function
to be generated from data — see
`handouts/03_chapter2_figures_needed.md`

Figure 2.8: Spike-timing dependent plasticity (STDP). Percentage conductance change ΔG as a function of the pre/post spike-timing delay Δt, with each branch fitted to the decaying exponential of eq. (2.3). Causal timing ($\Delta t < 0$) produces potentiation and anti-causal timing ($\Delta t > 0$) produces depression, with fitted time constant $\tau = 85$ ms–90 ms, within the biologically realistic range (~ 100 ms). Inset: pre- and postsynaptic voltage waveforms separated by Δt. Data from Ref. [17].

The result that the obtained learning rule corresponds to *asymmetric anti-Hebbian* STDP (rather than the symmetric Hebbian or asymmetric Hebbian variants also observed in biology) reflects the specific voltage waveform design and the polarity conventions of the electrode configuration. Different waveform designs would be expected to produce different STDP kernels, offering an additional degree of tunability in the design of hardware neural network architectures built from these devices.

2.5 Theoretical Framework: Ion Transport and the Origin of Memristance

2.5.1 Equilibrium Ion Distribution and Response to an Applied Field

In the absence of an applied voltage, the ions (Li^+ cations and $CF_3SO_3{}^-$ anions) are distributed homogeneously throughout the active layer. The equilibrium distribution is not strictly uniform at the atomic scale — each Li^+ ion occupies a local coordination site within the Hybrane matrix, and the triflate anions occupy interstitial positions between the coordinated cations — but on the scale of the film thickness (209 nm), the macroscopic charge density is uniform and the net electric field within the film is zero (apart from any built-in field arising from the electrode work function difference), as in the polymer-electrolyte picture inherited from LECs [27, 32, 46].

When a voltage is applied across the device, an electric field $\mathcal{E} = V/d$ (where d is the film thickness) is established. In the absence of free charge carriers that could screen this field, it would be uniform throughout the film. However, the mobile ions respond to the field by migrating: cations drift toward the cathode (ITO at positive bias on Ag), and anions drift toward the anode (Ag). This migration leads to the build-up of a charged double layer at each electrode interface on a timescale determined by the ionic mobility and the film thickness.

The resulting non-uniform ionic charge distribution modifies the internal electric field profile of the device significantly: the field is concentrated in a thin screening length near each electrode, while the bulk of the film is field-free [32, 47]. This spatial redistribution of field has significant consequences for charge injection, as described in the two competing models below.

2.5.2 Electrodynamic Model: Injection-Limited Transport

In the electrodynamic (ED) model, first formulated by deMello *et al.* [32, 47] to describe LECs, charge injection at the electrode–organic interface is considered to be the rate-limiting step for current flow. In this injection-limited regime, the current density is governed by thermionic emission over the Schottky barrier at the metal–polymer interface, described by:

$$J \propto T^2 \exp\left(-\frac{\Phi_B}{k_B T}\right) \tag{2.4}$$

where Φ_B is the effective barrier height, T is the absolute temperature, and k_B is the Boltzmann constant. The presence of the ionic double layer at the electrode interface reduces the effective barrier height Φ_B by bending the valence and conduction bands of the polymer: the concentration of positive ionic charge near the ITO electrode lowers the energy of the polymer conduction band edge at that interface, while the concentration of negative ionic charge near the Ag electrode raises the energy of the polymer valence band edge. In both cases, the result is a reduction of the effective injection barrier, increasing the current injection rate.

In the memristive context, the key implication of the ED model is that changes in conductance arise from changes in the interface ion concentration, which modifies Φ_B and hence the injection rate. The conductance is determined by the instantaneous ionic configuration near the electrodes: a higher concentration of positive ions near ITO means lower Φ_B at ITO, higher hole injection, and higher conductance. After the driving field is removed, the ions relax back toward their equilibrium distribution by thermal diffusion, restoring the original barrier height and conductance — this is the physical basis of the memory decay.

2.5.3 Electrochemical Doping Model: Ohmic Injection

The electrochemical doping (ECD) model, formulated by Pei *et al.* [27] and developed by subsequent authors [48, 49], considers instead that charge injection at the electrodes is ohmic (i.e., not rate-limiting). In this regime, the majority of the voltage drop occurs in the bulk of the film rather than at the interfaces, and the current is limited by the bulk conductivity. The defining feature of this model is that ion accumulation is sufficient to *electrochemically dope* the conjugated polymer: cations accumulate on the chains near the cathode, stabilising injected electrons and producing n-doped regions; anions accumulate near the anode, stabilising injected holes and producing p-doped regions. The doped regions have substantially higher conductivity than the undoped polymer, so the formation of the doped layers markedly increases the overall device conductance.

In the ECD model, the degree of doping — and hence the conductance — is determined by the extent of ionic migration, which is in turn governed by the history of applied voltage. This provides a natural mechanism for memristive behaviour: the conductance records the cumulative history of ionic displacement through the degree of polymer doping.

2.5.4 Resolution of the Two Models and the Role of Injection Regime

A prolonged debate in the LEC community regarding the relative validity of the ED and ECD models was resolved by van Reenen *et al.* [46], who showed that both models are correct in different limits. The ED model applies when the injection is strongly limited by the barrier (low-conductance organic semiconductor, large electrode work function mismatch); the ECD model applies when injection is ohmic (high-conductance semiconductor, small work function mismatch). In practice, a device may operate in the ED regime at early times (before significant ionic redistribution has occurred) and transition toward the ECD regime as the ionic double layers build up.

For the present SY/Hybrane/LiOTf system, the *Super Yellow* polymer has moderate hole mobility and a moderate injection barrier from ITO, so both mechanisms may contribute simultaneously. The interpretation adopted here associates the ED mechanism with the rapid conductance changes at short timescales (milliseconds to seconds), driven primarily by the fast-moving triflate anions, and the ECD mechanism with the larger, slower changes at longer timescales (seconds to tens of seconds), driven by the slower-moving Li^+ cations; on this reading the hierarchy of timescales is the physical origin of the STM/LTM dichotomy. It should be stressed that the device-level measurements of section 2.4 do not by themselves uniquely separate the two contributions: a clean discrimination would require, for instance, their temperature, film-thickness, or sweep-rate dependence, which is not pursued here. The assignment is therefore a working interpretation rather than an established fact. Its central falsifiable consequence — that the slow timescale should track the cation–host coordination strength — is examined directly in Chapter 3 by varying the cation at fixed host chemistry.

2.5.5 Lewis Acid-Base Chemistry and the Two Memory Regimes

The chemical mechanism proposed here for the two memory regimes is the same hard–soft acid–base reasoning developed in **????**, applied to the specific salt and host of this device.

Within the HSAB framework, the strength of the interaction between a Lewis acid and a Lewis base in a coordination environment is set by the matching of their hardness or softness [11]. The hard–hard interaction between Li^+ (a small, hard Lewis acid with charge density $z/r \approx 1.3\,\text{Å}^{-1}$–$1.7\,\text{Å}^{-1}$) and the ether oxygen atoms of *Hybrane* (hard Lewis bases) is governed predominantly by electrostatic forces, and is therefore thermodynamically strong and geometrically specific. In polyether environments Li^+ is coordinated by four to six oxygen donors — five in the crystalline complex PEO_3:$LiCF_3SO_3$ (three ether oxygens and two triflate oxygens) — with oxygen–lithium distances of 1.9 Å–2.1 Å [24]. This local site *binding* should not be confused with the *migration* activation barrier that governs hopping: the latter is the saddle-point barrier between adjacent sites — of order $50\,\mathrm{kJ\,mol^{-1}}$ ($\approx 20\,k_\mathrm{B}T$; section 2.2.3) — and is therefore smaller than the full coordination well, while still being substantially larger than $k_\mathrm{B}T$.

The interaction between the $CF_3SO_3^-$ triflate anion and the host oxygens is, by contrast, much weaker. The diffuse, low-charge-density triflate anion does not engage in strong electrostatic interactions with the hard-base oxygen atoms, and the residual interaction is governed mainly by weak van der Waals forces [50, 51]. The energy barrier for displacement of a triflate anion from one interstitial site to the next is therefore expected to be substantially lower than the barrier for a Li^+ cation.

This differential activation barrier translates directly into a differential ionic mobility under an applied field, and hence into different voltage thresholds for significant ionic redistribution:

- A modest voltage of 1 V applied across a 209 nm film generates a field of approximately 4.8 $\mathrm{MV\,m^{-1}}$. In the interpretation adopted here, this is sufficient to drive the more weakly bound triflate-anion redistribution observed experimentally [17]. The fast anionic redistribution produces a rapid conductance change, which then relaxes on the timescale of triflate diffusion through the host. This is the short-term memory regime, with retention 10 s–15 s, driven primarily by triflate relaxation.

- A voltage of 3 V across the same film generates approximately 14.4 $\mathrm{MV\,m^{-1}}$. This higher field is assigned to the onset of a slower cation contribution, consistent with the stronger Li^+ coordination expected in oxygen-rich polyether hosts [24, 50]. The cation contribution sets in, produces a delayed and stronger ionic redistribution, and relaxes on the much longer timescale of cation re-coordination. This is the long-term memory regime, with retention exceeding 45 s, driven primarily by cation redistribution.

This account yields a specific, experimentally testable prediction. Replacing Li^+ with a monovalent cation of larger ionic radius (Na^+ or K^+) should reduce the charge density, and therefore the hardness, of the Lewis acid. The cation–oxygen coordination with the polyether host should weaken accordingly, and the slow fading-memory timescale of the device should shorten as the cation is moved along the $Li^+ \rightarrow Na^+ \rightarrow K^+$ series. The prediction is tested in Chapter 3 on the main comparative corpus of the thesis: a PEO-based polymer-electrolyte series loaded with lithium, sodium and potassium triflate salts (PEO/LiOTf, PEO/NaOTf, PEO/KOTf) at fixed polymer and salt mass fractions, complemented by a dedicated PEO/LiOTf concentration sub-study. The prediction is addressed there as a statement about the shift of the measured variable-delay depotentiation timescale when the cation is varied at fixed host chemistry, not as a claim about the full synaptic suite reported in the present chapter.

2.6 Device Stability and Reproducibility

2.6.1 Ambient Shelf Stability

Two distinct notions of device longevity must be kept separate. *Cycling endurance* — the number of write/erase cycles a device sustains before its performance parameters drift outside specified tolerances — is the metric that ultimately governs suitability for the millions-to-billions of weight updates of on-chip training; it is not characterised for the present devices and is left for future work. *Shelf (calendar) stability* — the retention of performance during storage and intermittent operation under ambient conditions — is what is reported here. It is the more immediately relevant metric at the proof-of-concept stage, where the question is simply whether the unencapsulated composite survives realistic handling at all.

The devices reported here were characterised at regular intervals over a period of two weeks ($\sim$300 h) after fabrication, without encapsulation, while stored in ambient air at room temperature (see Supporting Information of Ref. [17]). The key metrics monitored were: the baseline conductance G_0, the maximum achievable conductance change $\Delta G_{\max}$, and the STM retention time τ_S.

The results show that all three metrics remain within 20 % of their initial values over the full two-week measurement period [17]. This stability compares favourably with other organic memristive devices reported in the literature, which typically exhibit significant degradation within hours to days under ambient conditions [52]. The enhanced stability is attributed to the solid-state nature of the composite: unlike devices based on liquid electrolytes or gel polymers, the Hybrane matrix provides a mechanically cohesive environment that protects the ions from evaporation and the polymer films from delamination, consistent with the stability of related Hybrane-based LEC blends [20].

The stability under ambient conditions is expected to be further enhanced by encapsulation, which is standard practice in organic electronic device manufacturing. Encapsulation with a thin film of SiO_2 or Si_3N_4 deposited by atomic layer deposition (ALD) or chemical vapour deposition (CVD) would eliminate oxygen and water ingress, the two primary degradation pathways [53]. The demonstration that even unencapsulated devices survive for two weeks validates the fundamental robustness of the composite material system.

2.6.2 Device-to-Device Reproducibility

Reproducibility is a persistent challenge for memristive devices, particularly those based on filamentary switching in inorganic materials, where the stochastic nucleation of conducting filaments produces large device-to-device and cycle-to-cycle variability [54]. The present system, which operates by uniform bulk ionic redistribution rather than localised filament formation, is intrinsically less susceptible to this type of variability.

Reproducibility was assessed by measuring the EPSC protocol and the potentiation/depression pulse sequence on multiple independent devices fabricated in the same batch. The relative standard deviation (RSD) of the EPSC ratios across devices is below 15 % for all four states (S_1 through S_4). The RSD of the ΔG values in the potentiation experiment is below 10 % for $N_{\text{pulses}} < 100$, as shown by the shaded error bands in fig. 2.5.

This level of reproducibility reflects the benefits of solution-processing over physical vapour deposition (PVD) for organic devices: the active layer is deposited from a well-mixed, homogeneous solution, so the composition and morphology of the film are governed by thermodynamic equilibrium rather than the kinetics of vapour phase deposition [13, 55]. Batch-to-batch reproducibility was also assessed and found to be within 20 % for the principal electrical parameters.

2.7 Comparison with the State of the Art

Table 2.1 places the device reported in this chapter alongside representative organic and hybrid memristive devices from the literature available at the time of publication (2022). The comparison axes are those of **??**: device structure (2-T or 3-T), accessible conductance states, demonstrated short- and long-term memory regimes, STDP, and reversibility.

Two combinations not previously demonstrated in a single platform stand out from the table. The first is the coexistence of short-term memory and long-term memory in a single 2-T device that is also fully reversible. Of the prior reports, Liu and co-workers [5] comes closest, having shown tunable short- to long-term transitions in a 2-T structure; the redox route, however, modifies the organic active layer irreversibly at each switching cycle, which precludes practical cyclic operation [6]. The present device reaches comparable short-term memory/long-term memory functionality with full reversibility (the conductance returns to G_0 after the reset protocol) and with the retention stability described in section 2.6. The second is the demonstration of STDP in a fully reversible 2-T architecture. The earlier organic STDP demonstrations [7, 12, 58] were all 3-T, with an independent gate voltage providing the timing-dependent rule; here, the same rule emerges from the superposition of pre- and postsynaptic pulses at the single Ag terminal, which is closer in form to the biological

Table 2.1: Comparison of key performance metrics of the present work with representative organic and hybrid memristive devices from the literature. "2-T" = two-terminal; "3-T" = three-terminal; "STM" = short-term memory; "LTM" = long-term memory; "STDP" = spike-timing dependent plasticity; "—" = not reported or not applicable.

Reference	**Material**	**Structure**	**States**	**STM**	**LTM**	**STDP**	**Reversible**
Zakhidov 2010 [4]	$[Ru(bpy)_3]^{2+}/PF_6^-$	2-T	2	—	—	No	Partial
Lei 2014 [56]	PVA film	2-T	2	—	—	No	No
Liu 2016 [5]	$[EV(ClO_4)]/(BTPA\text{-}F)$	2-T	cont.	Yes (tunable)	Yes	No	No
Gkoupidenis 2015 [7]	PEDOT:PSS (OECT)	3-T	cont.	Yes	No	Yes	Yes
Xu 2016 [8]	PEDOT:PSS (OECT)	3-T	cont.	Yes	Yes	No	Yes
Kim 2018 [12]	Elastic organic composite	3-T	cont.	Yes	No	Yes	Yes
Goswami 2017 [57]	Molecular switch	2-T	2	—	Yes	No	Yes
Seo 2019 [58]	P3HT organic FET	3-T	cont.	Yes	No	Yes	Yes
This work [17]	**SY/Hybrane/LiOTf**	**2-T**	**cont.**	**Yes (10–15 s)**	**Yes (>45 s)**	**Yes**	**Full**

synapse where no separate gating terminal is present.

The other axes summarise the operating regime of the device rather than a competitive claim. The ambient shelf lifetime (about 300 h unencapsulated) exceeds that of most comparable unencapsulated organic memristive devices by roughly an order of magnitude [52]; cycling endurance, a distinct metric not characterised here, is the axis on which mature inorganic platforms reach $> 10^{10}$ cycles [59]. The energy per synaptic event (about 50 nJ) is consistent with the range reported for other organic ionic devices [38]; it is, however, higher than the picojoule-scale energy per synaptic operation of dedicated CMOS neuromorphic hardware [60], in which even the elementary arithmetic and memory-access operations underlying a weight update already lie in the picojoule range [61], and it remains above the picojoule budget of the best inorganic memristors [39]. These differences reflect the lower switching speed and larger device area of this proof-of-concept organic system. Scaling the device area toward the $\sim$100 nm^2 regime considered in the inorganic literature would reduce this gap by several orders of magnitude, although such scaling is not pursued here.

The comparison above retains the device-level metrics conventional in the organic memristive literature of the early 2020s: reversibility, retention window, energy per event, and operational lifetime. Within the thesis-wide argument, those metrics play a narrower role. They establish that the polymer-electrolyte platform introduced here is reversible, dynamically addressable, and stable enough to support the comparative study of Chapter 3 (a PEO-based triflate composition and cation sweep, characterised through the same three common dynamical measurements) and the temporal-computing reinterpretation of Chapter 4 built on those measurements. They are not invoked to claim that this device should be judged by the standards of a high-density non-volatile memory technology, nor to suggest that the full synaptic suite reported here is available across the full comparative corpus. Those measurements remain specific to the present SY/Hybrane/LiOTf device.

2.8 Summary

This chapter has reported the design, fabrication, and characterisation of a fully reversible 2-T organic memristive device that supports, within a single polymer-composite active layer, a full set of synaptic functions organised in two families: the three dynamical measurements carried through the rest of the thesis (I–V hysteresis, variable-number-of-pulses potentiation, and variable-delay depotentiation), and the extended synaptic protocols specific to this device (EPSC, the separated short-term memory/long-term memory reading of the retention data, and STDP). The main contributions are as follows.

1. **Materials design.** A ternary composite of *Super Yellow* (semiconducting polymer), *Hybrane* (ion-conducting hyperbranched polyester-amide host), and LiOTf (alkali-metal salt) was shown to satisfy the three functional requirements identified in section 2.2.1: electronic-conductance pathway, ion-transport medium, and ion source. The composition follows the composite logic developed in **??**.

2. **Device fabrication.** A 2-T vertical sandwich (ITO/composite/Ag) was fabricated from solution processing and thermal evaporation alone, yielding active layers of 209 nm thickness with root-mean-square surface roughness below 3 nm and batch-to-batch reproducibility within 20 % on the principal electrical parameters.

3. **Current–voltage hysteresis.** The first of the three dynamical measurements carried forward to the comparative corpus: successive triangular voltage sweeps produced a continuously increasing

conductance, with a 140 % increase from cycle 1 to cycle 2 at 1 V — the analogue, non-abrupt signature of memristive switching.

4. **Variable-number-of-pulses potentiation.** Pulse-train measurements confirmed that the conductance can be potentiated and depotentiated with full reversibility over a range exceeding 200 %, at an energy cost of order 50 nJ per event, and that the conductance change grows monotonically with the number of write pulses, exceeding 10^3% for $N_{\mathrm{pulses}} > 10^3$.

5. **Variable-delay depotentiation (short-term memory and long-term memory).** Read-out after a variable delay resolved two distinct memory regimes: an short-term memory regime ($\tau_S =$ 2.5 s–3 s, retention 10 s–15 s) accessible with 1 V pulses, and a longer-lived regime denoted long-term memory ($\tau_L = 4.7$ s, retention > 45 s) accessible with 3 V pulses. Both decay curves are well described by a stretched Kohlrausch exponential, in keeping with a heterogeneous distribution of activation barriers within the polymer matrix.

6. **EPSC multi-state readout.** The first of the extended protocols specific to this device: four readily distinguishable readout states were demonstrated around the device ground level, with signed EPSC readout ratios $R_n = I_{S_n}/I_{S_0}$ spanning from −17 to +16 relative to the ground-state response S_0 under the fixed read-out bias; given the analogue conductance tunability, this is a representative demonstration rather than an intrinsic limit on the number of accessible states.

7. **STDP.** An asymmetric anti-Hebbian STDP function was measured, with time constant $\tau = 85$ ms–90 ms, within the biologically realistic range of ~ 100 ms [14, 15].

8. **Mechanistic account.** The short-term memory/long-term memory dichotomy was attributed to the differential Lewis acid–base

interaction between the mobile ions (Li^+ as hard acid, $CF_3SO_3^-$ as soft base) and the host oxygen atoms of *Hybrane* (hard Lewis bases). The weakly coordinated triflate anion is displaced at low bias and relaxes on a short timescale, giving rise to the short-term memory regime. The strongly coordinated lithium cation requires a higher bias to leave its coordination site and relaxes on a longer timescale, giving rise to the long-term memory regime. The injection and bulk-transport physics are described jointly by the electrodynamic and electrochemical-doping models, in the reconciling framework of van Reenen and co-workers [46].

9. **Stability and reproducibility.** Devices retained their performance over an ambient shelf lifetime of approximately 300 h without encapsulation (cycling endurance, a distinct metric, was not characterised), with device-to-device variability below 15 % on the principal electrical parameters.

Taken together, these results establish the SY/Hybrane/LiOTf composite as a thoroughly characterised exemplar of a polymer-electrolyte organic memristive device, and identify cation–polyether coordination chemistry as the physical hypothesis for memory-timescale control. This identification motivates the main comparative study of Chapter 3, which replaces *Hybrane* with poly(ethylene oxide), varies the PEO/LiOTf composition as the replicated quantitative axis, and surveys host, anion, and cation substitutions as sample-limited chemical side evidence. In Chapter 3, the hypothesis is tested through the three common dynamical measurements (I–V hysteresis, variable-number-of-pulses potentiation, and variable-delay depotentiation) that are abundant across the comparative corpus; the outcome is deliberately cautious, with no robust host- and anion-independent Li/Na/K law claimed. The richer synaptic metrics reported in the present chapter (EPSC, separated short-term memory/long-term memory retention, and STDP) are not propagated to the later chapters and serve there only as lithium-device priors and sanity checks. Beyond this comparative handoff, the volatile two-timescale behaviour established here is the experimental anchor for the temporal-computing reinterpretation of Chapter 4, in which behavioural models fitted to the Chapter 3 corpus drive reservoir-computing simulations, physiological temporal-context reconstruction, and WESAD affective-computing demonstrations. In those paradigms, the fading memory of the device is the active computational resource rather than a retention defect to be suppressed.

References

1. Erokhin, V., Berzina, T. & Fontana, M. P. Hybrid electronic device based on polyaniline-polyethyleneoxide junction. *Journal of Applied Physics* **97,** 064501. doi:10.1063/1.1861508 (2005).

2. Smerieri, A., Berzina, T., Erokhin, V. & Fontana, M. P. Polymeric electrochemical element for adaptive networks: Pulse mode. *Journal of Applied Physics* **104,** 114513. doi:10.1063/1.3033399 (2008).

3. Berzina, T., Smerieri, A., Bernabò, M., *et al.* Optimization of an organic memristor as an adaptive memory element. *Journal of Applied Physics* **105,** 124515. doi:10.1063/1.3153944 (2009).

4. Zakhidov, A. A., Jung, B., Slinker, J. D., Abruña, H. D. & Malliaras, G. G. A light-emitting memristor. *Organic Electronics* **11,** 150–153. doi:10.1016/j.orgel.2009.09.015 (2010).

5. Liu, G., Wang, C., Zhang, W., *et al.* Organic Biomimicking Memristor for Information Storage and Processing Applications. *Advanced Electronic Materials* **2,** 1500298. doi:10.1002/aelm.201500298 (2016).

6. Heremans, P., Gelinck, G. H., Müller, R., Baeg, K.-J., Kim, D.-Y. & Noh, Y.-Y. Polymer and Organic Nonvolatile Memory Devices. *Chemistry of Materials* **23,** 341–358. doi:10.1021/cm102006v (2011).

7. Gkoupidenis, P., Schaefer, N., Garlan, B. & Malliaras, G. G. Neuromorphic Functions in PEDOT:PSS Organic Electrochemical Transistors. *Advanced Materials* **27,** 7176–7180. doi:10.1002/adma.201503674 (2015).

8. Xu, W., Min, S.-Y., Hwang, H. & Lee, T.-W. Organic core-sheath nanowire artificial synapses with femtojoule energy consumption. *Science Advances* **2,** e1501326. doi:10.1126/sciadv.1501326 (2016).

9. Van de Burgt, Y., Lubberman, E., Fuller, E. J., *et al.* A non-volatile organic electrochemical device as a low-voltage artificial synapse for neuromorphic computing. *Nature Materials* **16,** 414–418. doi:10.1038/nmat4856 (2017).

10. Fuller, E. J., Keene, S. T., Melianas, A., *et al.* Parallel programming of an ionic floating-gate memory array for scalable neuromorphic computing. *Science* **364,** 570–574. doi:10.1126/science.aaw5581 (2019).

11. Pearson, R. G. Hard and Soft Acids and Bases. *Journal of the American Chemical Society* **85,** 3533–3539. doi:10.1021/ja00905a001 (1963).

12. Kim, Y., Chortos, A., Xu, W., *et al.* A bioinspired flexible organic artificial afferent nerve. *Science* **360,** 998–1003. doi:10.1126/science.aao0098 (2018).

13. Van de Burgt, Y., Melianas, A., Keene, S. T., Malliaras, G. & Salleo, A. Organic electronics for neuromorphic computing. *Nature Electronics* **1,** 386–397. doi:10.1038/s41928-018-0103-3 (2018).

14. Bi, G.-q. & Poo, M.-m. Synaptic Modifications in Cultured Hippocampal Neurons: Dependence on Spike Timing, Synaptic Strength, and Postsynaptic Cell Type. *Journal of Neuroscience* **18,** 10464–10472. doi:10.1523/JNEUROSCI.18-24-10464.1998 (1998).

15. Kuzum, D., Yu, S. & Wong, H.-S. P. Synaptic electronics: materials, devices and applications. *Nanotechnology* **24,** 382001. doi:10.1088/0957-4484/24/38/382001 (2013).

16. Shannon, R. D. Revised effective ionic radii and systematic studies of interatomic distances in halides and chalcogenides. *Acta Crystallographica Section A* **32,** 751–767. doi:10.1107/S0567739476001551 (1976).

17. Prado-Socorro, C. D., Giménez-Santamarina, S., Mardegan, L., *et al.* Polymer-Based Composites for Engineering Organic Memristive Devices. *Advanced Electronic Materials* **8,** 2101192. doi:10.1002/aelm.202101192 (2022).

18. Zhang, M., Höfle, S., Czolk, J., Mertens, A. & Colsmann, A. All-solution processed transparent organic light emitting diodes. *Nanoscale* **7,** 20009–20014. doi:10.1039/C5NR05820A (2015).

19. Friend, R. H., Gymer, R. W., Holmes, A. B., *et al.* Electroluminescence in conjugated polymers. *Nature* **397,** 121–128. doi:10.1038/16393 (1999).

20. Mardegan, L., Dreessen, C., Sessolo, M., Tordera, D. & Bolink, H. J. Stable Light-Emitting Electrochemical Cells Using Hyperbranched Polymer Electrolyte. *Advanced Functional Materials* **31,** 2104249. doi:10.1002/adfm.202104249 (2021).

21. Froehling, P. Development of DSM's Hybrane hyperbranched polyesteramides. *Journal of Polymer Science Part A: Polymer Chemistry* **42,** 3110–3115. doi:10.1002/pola.20113 (2004).

22. Armand, M. Polymer solid electrolytes – an overview. *Solid State Ionics* **9-10,** 745–754. doi:10.1016/0167-2738(83)90083-8 (1983).

23. Berthier, C., Gorecki, W., Minier, M., Armand, M. B., Chabagno, J. M. & Rigaud, P. Microscopic investigation of ionic conductivity in alkali metal salts-poly(ethylene oxide) adducts. *Solid State Ionics* **11,** 91–95. doi:10.1016/0167-2738(83)90068-1 (1983).

24. Lightfoot, P., Mehta, M. A. & Bruce, P. G. Crystal Structure of the Polymer Electrolyte Poly(ethylene oxide)3:LiCF3SO3. *Science* **262,** 883–885. doi:10.1126/science.262.5135.883 (1993).

25. Williams, M. L., Landel, R. F. & Ferry, J. D. The Temperature Dependence of Relaxation Mechanisms in Amorphous Polymers and Other Glass-forming Liquids. *Journal of the American Chemical Society* **77,** 3701–3707. doi:10.1021/ja01619a008 (1955).

26. Gray, F. M. *Polymer Electrolytes* (Royal Society of Chemistry, Cambridge, 1997).

27. Pei, Q., Yu, G., Zhang, C., Yang, Y. & Heeger, A. J. Polymer Light-Emitting Electrochemical Cells. *Science* **269,** 1086–1088. doi:10.1126/science.269.5227.1086 (1995).

28. Wågberg, T., Hania, P. R., Robinson, N. D., Shin, J.-H., Matyba, P. & Edman, L. On the Limited Operational Lifetime of Light-Emitting Electrochemical Cells. *Advanced Materials* **20,** 1744–1749. doi:10.1002/adma.200702595 (2008).

29. Gkoupidenis, P., Koutsouras, D. A. & Malliaras, G. G. Neuromorphic device architectures with global connectivity through electrolyte gating. *Nature Communications* **8,** 15448. doi:10.1038/ncomms15448 (2017).

30. Kim, H., Gilmore, C. M., Piqué, A., *et al.* Electrical, optical, and structural properties of indium–tin–oxide thin films for organic light-emitting devices. *Journal of Applied Physics* **86,** 6451–6461. doi:10.1063/1.371708 (1999).

31. Ramsdale, C. M., Barker, J. A., Arias, A. C., MacKenzie, J. D., Friend, R. H. & Greenham, N. C. The origin of the open-circuit voltage in polyfluorene-based photovoltaic devices. *Journal of Applied Physics* **92,** 4266–4270. doi:10.1063/1.1506385 (2002).

32. De Mello, J. C., Tessler, N., Graham, S. C. & Friend, R. H. Ionic space-charge effects in polymer light-emitting diodes. *Physical Review B* **57,** 12951–12963. doi:10.1103/PhysRevB.57.12951 (1998).

33. Hung, L. S., Tang, C. W. & Mason, M. G. Enhanced electron injection in organic electroluminescence devices using an Al/LiF electrode. *Applied Physics Letters* **70,** 152–154. doi:10.1063/1.118344 (1997).

34. Chua, L. O. Memristor – The Missing Circuit Element. *IEEE Transactions on Circuit Theory* **18,** 507–519. doi:10.1109/TCT.1971.1083337 (1971).

35. Adhikari, S. P., Sah, M. P., Kim, H. & Chua, L. O. Three Fingerprints of Memristor. *IEEE Transactions on Circuits and Systems I: Regular Papers* **60,** 3008–3021. doi:10.1109/TCSI.2013.2256171 (2013).

36. Chua, L. O. If it's pinched it's a memristor. *Semiconductor Science and Technology* **29,** 104001. doi:10.1088/0268-1242/29/10/104001 (2014).

37. Fang, J., Matyba, P., Robinson, N. D. & Edman, L. Identifying and alleviating electrochemical side-reactions in light-emitting electrochemical cells. *Journal of the American Chemical Society* **130,** 4562–4568. doi:10.1021/ja7113294 (2008).

38. Xiao, Z. & Huang, J. Energy-Efficient Hybrid Perovskite Memristors and Synaptic Devices. *Advanced Electronic Materials* **2,** 1600100. doi:10.1002/aelm.201600100 (2016).

39. Yang, J. J., Strukov, D. B. & Stewart, D. R. Memristive devices for computing. *Nature Nanotechnology* **8,** 13–24. doi:10.1038/nnano.2012.240 (2013).

40. Zucker, R. S. Short-term synaptic plasticity. *Annual Review of Neuroscience* **12,** 13–31. doi:10.1146/annurev.ne.12.030189.000305 (1989).

41. Zucker, R. S. & Regehr, W. G. Short-term synaptic plasticity. *Annual Review of Physiology* **64,** 355–405. doi:10.1146/annurev.physiol.64.092501.114547 (2002).

42. Sturman, B., Podivilov, E. & Gorkunov, M. Origin of Stretched Exponential Relaxation for Hopping-Transport Models. *Physical Review Letters* **91,** 176602. doi:10.1103/PhysRevLett.91.176602 (2003).

43. Hebb, D. O. *The Organization of Behavior: A Neuropsychological Theory* (Wiley, New York, 1949).

44. Markram, H., Lübke, J., Frotscher, M. & Sakmann, B. Regulation of synaptic efficacy by coincidence of postsynaptic APs and EPSPs. *Science* **275,** 213–215. doi:10.1126/science.275.5297.213 (1997).

45. Bichler, O., Querlioz, D., Thorpe, S. J., Bourgoin, J.-P. & Gamrat, C. Extraction of temporally correlated features from dynamic vision sensors with spike-timing-dependent plasticity. *Neural Networks* **32,** 339–348. doi:10.1016/j.neunet.2012.02.022 (2012).

46. Van Reenen, S., Matyba, P., Dzwilewski, A., Janssen, R. A. J., Edman, L. & Kemerink, M. A Unifying Model for the Operation of Light-Emitting Electrochemical Cells. *Journal of the American Chemical Society* **132,** 13776–13781. doi:10.1021/ja1045555 (2010).

47. De Mello, J. C. Interfacial feedback dynamics in polymer light-emitting electrochemical cells. *Physical Review B* **66,** 235210. doi:10.1103/PhysRevB.66.235210 (2002).

48. Smith, D. L. Steady state model for polymer light-emitting electrochemical cells. *Journal of Applied Physics* **81,** 2869–2880. doi:10.1063/1.363966 (1997).

49. Matyba, P., Maturova, K., Kemerink, M., Robinson, N. D. & Edman, L. The dynamic organic p-n junction. *Nature Materials* **8,** 672–676. doi:10.1038/nmat2478 (2009).

50. Bruce, P. G. & Vincent, C. A. Polymer electrolytes. *Journal of the Chemical Society, Faraday Transactions* **89,** 3187–3203. doi:10.1039/ft9938903187 (1993).

51. Frech, R., Chintapalli, S., Bruce, P. G. & Vincent, C. A. Crystalline and Amorphous Phases in the Poly(ethylene oxide)–LiCF3SO3 System. *Macromolecules* **32,** 808–813. doi:10.1021/ma9812682 (1999).

52. Shih, C.-C., Lee, W.-Y. & Chen, W.-C. Nanostructured materials for non-volatile organic transistor memory applications. *Materials Horizons* **3,** 294–308. doi:10.1039/C6MH00049E (2016).

53. Lewis, J. S. & Weaver, M. S. Thin-Film Permeation-Barrier Technology for Flexible Organic Light-Emitting Devices. *IEEE Journal of Selected Topics in Quantum Electronics* **10,** 45–57. doi:10.1109/JSTQE.2004.824072 (2004).

54. Sun, W., Gao, B., Chi, M., *et al.* Understanding memristive switching via in situ characterization and device modeling. *Nature Communications* **10,** 3453. doi:10.1038/s41467-019-11411-6 (2019).

55. Sangwan, V. K. & Hersam, M. C. Neuromorphic nanoelectronic materials. *Nature Nanotechnology* **15,** 517–528. doi:10.1038/s41565-020-0647-z (2020).

56. Lei, Y., Liu, Y., Xia, Y., *et al.* Memristive learning and memory functions in polyvinyl alcohol polymer memristors. *AIP Advances* **4,** 077105. doi:10.1063/1.4887010 (2014).

57. Goswami, S., Matula, A. J., Rath, S. P., *et al.* Robust resistive memory devices using solution-processable metal-coordinated azo aromatics. *Nature Materials* **16,** 1216–1224. doi:10.1038/nmat5009 (2017).

58. Seo, D.-G., Lee, Y., Go, G.-T., *et al.* Versatile neuromorphic electronics by modulating synaptic decay of single organic synaptic transistor: From artificial neural networks to neuro-prosthetics. *Nano Energy* **65,** 104035. doi:10.1016/j.nanoen.2019.104035 (2019).

59. Ielmini, D. & Wong, H.-S. P. In-memory computing with resistive switching devices. *Nature Electronics* **1,** 333–343. doi:10.1038/s41928-018-0092-2 (2018).

60. Indiveri, G. & Liu, S.-C. Memory and Information Processing in Neuromorphic Systems. *Proceedings of the IEEE* **103,** 1379–1397. doi:10.1109/JPROC.2015.2444094 (2015).

61. Horowitz, M. *1.1 Computing's energy problem (and what we can do about it)* in *2014 IEEE International Solid-State Circuits Conference Digest of Technical Papers (ISSCC)* (2014), 10–14. doi:10.1109/ISSCC.2014.6757323.